

Extended Kalman Filter Tuning in Sensorless PMSM Drives

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Abstract

The use of an Extended Kalman Filter (EKF) as non-linear speed and position observer for Permanent Magnet Synchronous Motor (PMSM) drives is a mature research topic. Notwithstanding, the shift from research prototype to a market-ready product still calls for a solution of some implementation pitfalls. The major and still unsolved topic is the choice of the EKF covariance matrices. This paper replaces the usual trial-and-error method with a straightforward matrices choice. These matrices, possibly combined with a novel self-tuning procedure, should put the EKF drive much closer to an off-the-shelf product. The proposed method is based on the complete normalisation of the EKF algorithm representation. Successful experimental results are included in the paper.

1 Introduction

The Extended Kalman Filter (EKF) is an optimal estimator in the least-square sense for estimating the states of dynamic non-linear systems, and it is thus a viable and computationally efficient candidate for the on-line determination of rotor position and speed of a Permanent Magnet Synchronous Motor (PMSM) [1]-[4]. Theoretical basis and digital implementation of EKF have been deeply investigated [2],[3],[5]. However, at least one major drawback of the EKF application to sensorless drives is still unsolved, that is the design and the tuning of the covariance matrices that appear in the EKF equations.

Covariance matrices account for model approximation and measurement noise. They can be obtained by considering the stochastic properties of the corresponding noises. Since these are usually unknown, in most cases the EKF matrices are designed and tuned by trial and error procedures. Skilled personnel varies the matrix elements in a range of several decades, in order to get the best fit for the specific application. Indeed, it is the customisation required by each application that makes most of the EKF-based drives incompatible with an off-the-shelf market strategy.

This paper proposes a novel approach to the EKF settings for sensorless PMSM drives on the perspective of an industrial application. The idea has born from the

consideration of both drive control and motor electromechanical design aspects. It is first worth to note that by adopting a suitable normalisation, the PMSM parameters with isotropic rotor vary in a narrow range, regardless of motor size. If a coherent normalisation of the EKF algorithm is accomplished as well, the covariance matrices of the filter would generally fit for almost all standard PMSM drives. This smoothes the way towards an off-the-shelf oriented production, with the related benefits in the drives market place.

Of course, optimal sensorless drive performances can be achieved by a further on the field fine-tuning of the EKF covariance matrices. But it is worth to highlight that, opposite to the actual practice, the proposed method gives an effective and general initial guess for EKF matrices settings. Eventually, it will be proved that on-the-field refinements could be obtained by a particular application of the well-known Bartlett test, performed at first start-up of the drive.

The validity and the generality of the proposed EKF setting method have been tested by experiments on different laboratory prototypes.

2 Normalised Drive Equations

2.1 Motor Equations

Normalised equations are obtained by dividing each variable of interest by the corresponding base value.

Primary base quantities are the torque T_B , the electrical speed Ω_B and the maximum phase voltage U_B at the maximum power point of the constant torque region. Derived base quantities, such as motor parameters and current base values are fixed by equating the electrical power $(3/2)I_B U_B$ to the mechanical one $T_B \Omega_B / p$, where p denotes the number of pole pairs. The complete set of base values is reported in Table 1.

It is worth to note that the base value for the position is $\theta_B = 1$. Angular position is thus the only quantity unaffected by the normalisation, that is to say, every sinusoidal quantity has a repetition period of 2π . In the following, normalised quantities will be denoted by the superscript "n".

Table 1. Base values for the normalisation

Base quantity	Symbol	Value
Current	I_B	$2T_B\Omega_B/(3pU_B)$
Flux	Λ_B	U_B/Ω_B
Inductance	L_B	Λ_B/I_B
Resistance	R_B	U_B/I_B
Viscous coeff.	B_B	pT_B/Ω_B
Inertia	J_B	pB_B/Ω
Angle	ϑ_B	1 rad
Time	t_B	$1/\Omega_B$

Let's assume a stationary orthogonal reference frame (α, β) fixed to the stator. The two stator currents, the electrical speed and position are used as system state variables:

$$\mathbf{x} = [i_\alpha \ i_\beta \ \omega \ \vartheta]^\top \quad (1)$$

By defining the vector of system base states

$$\mathbf{x}_B = [I_B \ I_B \ \Omega_B \ \vartheta_B]^\top \quad (2)$$

the i -th element of the normalised state vector $\mathbf{x}^n$ can be simply obtained by

$$x_i^n = \frac{x_i}{x_{Bi}} \quad i = 1..4 \quad (3)$$

with obvious meaning of symbols. The system input and output are the phase voltages and current respectively:

$$\begin{aligned} \mathbf{u}^n &= [u_\alpha^n \ u_\beta^n]^\top \\ \mathbf{y}^n &= [i_\alpha^n \ i_\beta^n]^\top \end{aligned} \quad (4)$$

With the above assumptions and for a better matching to the EKF equations, the PMSM model equations can be rearranged in a canonical state-space form as

$$\begin{aligned} \dot{\mathbf{x}}^n &= \mathbf{f}(\mathbf{x}^n, \mathbf{u}^n) + \mathbf{v}^n \\ \mathbf{y}^n &= \mathbf{H}\mathbf{x}^n + \boldsymbol{\mu}^n \end{aligned} \quad (5)$$

where bold characters are used for vectors and matrices.

The normalised model and measurement disturbances are statistically described by the zero-mean Gaussian noises $\mathbf{v}^n(t)$ and $\boldsymbol{\mu}^n(t)$, whose variance matrices are $\mathbf{Q}^n$ and $\mathbf{R}^n$ respectively.

The functional $\mathbf{f}$ is given by

$$\mathbf{f} = \begin{bmatrix} \Omega_B \left(-\frac{R_s^n}{L_s^n} i_\alpha^n + \frac{\Lambda_{mg}^n \omega^n}{L_s^n} \sin(\vartheta^n) + \frac{u_\alpha^n}{L_s^n} \right) \\ \Omega_B \left(-\frac{R_s^n}{L_s^n} i_\beta^n - \frac{\Lambda_{mg}^n \omega^n}{L_s^n} \cos(\vartheta^n) + \frac{u_\beta^n}{L_s^n} \right) \\ 0 \\ \Omega_B \omega^n \end{bmatrix} \quad (6)$$

where R_s^n , L_s^n are the normalised motor resistance and inductance.

By imposing that the normalised torque, speed and voltage are all unity during operation at the maximum power point of the constant torque region, the normalised motor current is given by

$$I^n = \frac{1}{\Lambda_{mg}^n} \quad (7)$$

and the following relationship arises between stator inductance and PM flux linkage

$$L_s^n = \Lambda_{mg}^n \sqrt{1 - (\Lambda_{mg}^n)^2} \quad (8)$$

The developed torque is due to both the PM flux linkage and the stator current. In order to limit the latter, the PM flux linkage is normally designed as close as possible to its highest feasible value.

By considering several typical PMSM designs, it can be noted that the normalised parameters usually vary in the ranges

$$\Lambda_{mg}^n = 1 \div 0.9 \quad L_s^n = 0 \div 0.4 \quad I^n = 1 \div 1.1 \quad (9)$$

It is also worth to note that the leftmost boundary is only theoretical, since it refers to the ideal case of unity normalised PM flux linkage that in turn corresponds to unity power factor.

It is very important to highlight that the narrow ranges exhibited by the normalised motor parameters in (9) will make the EKF matrices largely independent of different motors parameter sets, resulting in matrices $\mathbf{Q}^n$, $\mathbf{R}^n$ of widespread validity.

2.2 Extended Kalman Filter Equations

The model described by (5) is formally identical to the motor model before normalisation. Therefore, the EKF algorithm [2], [3] can be straightforwardly applied, by manipulating normalised quantities.

As it comes clear from (6), the dynamic model of the motor is derived under the so-called "infinite inertia" hypothesis, that is the speed variation within each control cycle T_c is considered negligible with respect to the other variables dynamics. With this assumption, any mechanical load parameter, as well as the load torque, disappear from the equations of the model, for the sake of algorithm robustness.

For a correct digital implementation, the system (5) is first linearised around the best available estimate $\mathbf{x}e^n_o$ [3]. Let $\mathbf{F}$, $\mathbf{B}$ and $\mathbf{H}$ be respectively the state, the input and the output matrices of the linearised system.

At time $t_k = kT_c$ the optimal state estimate $\mathbf{x}e^n_{k|k}$ and the estimation error covariance matrix $\mathbf{P}^n_{k|k}$ are obtained through a simplified version of the EKF algorithm [3], summarised in Table 2.

Table 2. EKF algorithm

Prediction step
$\mathbf{x}_{k k-1}^n = \mathbf{x}_{k-1 k-1}^n + [\mathbf{f}(\mathbf{x}_{k-1 k-1}^n) + \mathbf{B} < \mathbf{u}_{k-1}^n >] T_c$
$\mathbf{P}_{k k-1}^n = \mathbf{P}_{k-1 k-1}^n + (\mathbf{F}_{k-1} \mathbf{P}_{k-1 k-1}^n \mathbf{F}_{k-1}^T + \mathbf{P}_{k-1 k-1}^n \mathbf{F}_{k-1}^T) T_c + \mathbf{Q}^n$
Innovation step
$\mathbf{x}_{k k}^n = \mathbf{x}_{k k-1}^n + \mathbf{K}_k^n [\mathbf{y}_k^n - \mathbf{H} \mathbf{x}_{k k-1}^n]$
$\mathbf{P}_{k k}^n = \mathbf{P}_{k k-1}^n - \mathbf{K}_k^n \mathbf{H} \mathbf{P}_{k k-1}^n$
Kalman gain
$\mathbf{K}_k^n = \mathbf{P}_{k k-1}^n \mathbf{H}' (\mathbf{H} \mathbf{P}_{k k-1}^n \mathbf{H}' + \mathbf{R}^n)^{-1}$

It is possible to demonstrate that between natural and normalised values for the EKF covariance matrices the following relations hold:

$$p_{ij}^n = \frac{P_{ij}}{x_{Bi} x_{Bj}} \quad q_{ii}^n = \frac{Q_{ii}}{(x_{Bi})^2} \quad r_{ii}^n = \frac{R_{ii}}{(x_{Bi})^2} \quad (10)$$

where $i, j \in [1..4]$ and x_{Bi} is the i -th element of system base states vector (2).

3 Drive Structure

The structure of the PMSM sensorless drive proposed in this paper is shown in Fig. 1, where normalised system has been put in evidence.

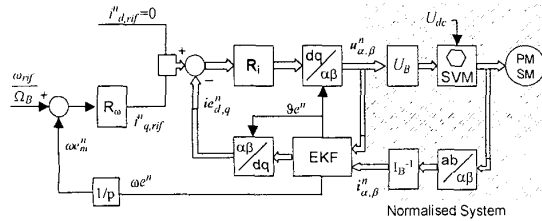


Fig. 1 - Sensorless PMSM drive with normalisation

An accurate compensation of any non-linearity of the Space Vector Modulation (SVM) inverter has been carried out. This makes possible the use of voltage references instead of measured voltages, with further reduction of the number of sensors.

Both speed and current loops have a control cycle $T_c = 125 \mu s$, as it is a satisfactory trade-off between algorithm efficiency and microprocessor load factor.

Current control is performed in a (d,q) synchronous coordinates system, fixed to the rotor PM. Two identical Proportional-Integral (PI) regulators have been designed and tuned to get a bandwidth $f_{oi} = \omega_{oi}/(2\pi) = 400$ Hz. Similarly, PI speed regulator's parameters have been set considering a bandwidth $f_{o\omega} = \omega_{o\omega}/(2\pi) = 20$ Hz.

PI constants are related to both the system parameters and the required bandwidth. Moreover, proper determination of the normalized values of the proportional and integral action for speed loop has

required an approximate knowledge of the total friction and inertia. For the experiments, the motor and load together featured $J^n = 11.32$ and $B^n = 0.05$.

The normalised expressions of all selected PI parameters are reported in Table 3.

Table 3. Normalised PI regulators

PI Laplace definition
$PI^n(s) = K_p^n + \frac{K_i^n}{s}$
Proportional Gain (Current Loop, subscript "i")
$K_{p,i}^n = \frac{K_{p,i}}{R_B} = \omega_{oi}^n L_s^n \sqrt{1 + (\omega_{oi}^n T_c^n)^2} = 0.5$
Integral Gain
$K_{i,i}^n = \frac{K_{i,i}}{R_B \Omega_B} = K_{p,i}^n \frac{R_s^n}{L_s^n} = 0.128$
Proportional Gain (Speed Loop, subscript "ω")
$K_{p,\omega}^n = \frac{K_{p,\omega}}{R_B} = \frac{2\omega_{o\omega}^n J^n \sqrt{1 + (\omega_{o\omega}^n / \omega_{oi}^n)^2}}{3pA_{mg}^n} = 3.11 \times 10^{-3}$
Integral Gain
$K_{i,\omega}^n = \frac{K_{i,\omega}}{R_B \Omega_B} = K_{p,\omega}^n \frac{B^n}{J^n} = 3.43 \times 10^{-6}$

The normalisation of each regulator constant has been performed by considering the physical dimension of the constant itself. For example, proportional gain is supposed to produce a voltage reference having a current error as input, and therefore it is similar to a resistance. This is the reason why the normalisation is carried out by the division by R_B .

4 Design and Tuning of Q and R

A critical point during the design of the EKF is the choice of the elements of the covariance matrices $\mathbf{P}_0$, $\mathbf{Q}$ and $\mathbf{R}$, which affect the performance and the convergence as well.

The diagonal initial state covariance matrix $\mathbf{P}_0$ represents variances or mean squared errors in the knowledge of the initial conditions. Varying $\mathbf{P}_0$ yields different amplitude of the transient, while both transient duration and steady state conditions will be unaffected [2]. Matrix $\mathbf{Q}$ gives the statistical description of the drive model. Increasing $\mathbf{Q}$ would indicate the presence of either heavy system noise or increased parameter uncertainty. An increment of the elements of $\mathbf{Q}$ will likewise increase the EKF gain, resulting in a faster filter dynamic. On the other hand, matrix $\mathbf{R}$ is related to measurement noise. Increasing the values of the elements of $\mathbf{R}$ will mean that the measurements are affected by noise and thus they are of little confidence.

Consequently the filter gain $\mathbf{K}$ will decrease, yielding poorer transient response.

It is a common practice to assume the covariance matrices $\mathbf{Q}$, $\mathbf{R}$ and $\mathbf{P}_0$ to be diagonal, for the lack of sufficient statistical information to evaluate their off-diagonal terms [1]-[3]. On the other hand, practice has revealed that even starting with non-zero off-diagonal values, at steady state off-diagonal terms remains several times smaller than the corresponding diagonal terms.

It can be realised that both $\mathbf{Q}$ and $\mathbf{R}$ depend on the drive parameters, the sampling time, the measurements amplitude and some other secondary factors.

The advantage of the proposed approach is that the appropriate design and tuning of $\mathbf{Q}$ and $\mathbf{R}$ made for a PMSM drive works well for any other drive with similar sampling time. It has also been experienced that the EKF behaviour is not influenced by different SVM technique.

Simulations and experimental tests on a drive prototype can derive appropriate ranges of normalized elements of the covariance matrices.

The first step of matrices tuning is to consider an open loop drive in which speed and current loops are closed around measured quantities. It is found that, without modifying PI regulators parameters, the effect of covariance elements variation is clear. Owing to negligence of the mechanical equation in the Kalman filter model, the estimated speed always delays the actual speed. This delay depends on the model covariance matrix $\mathbf{Q}$. Taking into account the behaviour of the open loop EKF to a speed reference step, it is found that the lower normalized elements $q_{1,1}^n = q_{2,2}^n$ of $\mathbf{Q}^n$, the lower the delay of the estimated speed in comparison with the effective one.

The aim of the first batch of experiments was devoted to find out the range of values that reduces the delay of the EKF response, being careful that small values of $q_{1,1}^n$ mean a high reliability of the motor model and this could not be verify by the experimental drive. Fig.2 shows actual and estimated speed step response for different values of $q_{1,1}^n$, with control loops closed on measured quantities, and by fixing $q_{3,3}^n = 4.6 \cdot 10^{-4}$, $q_{4,4}^n = 0.1$.

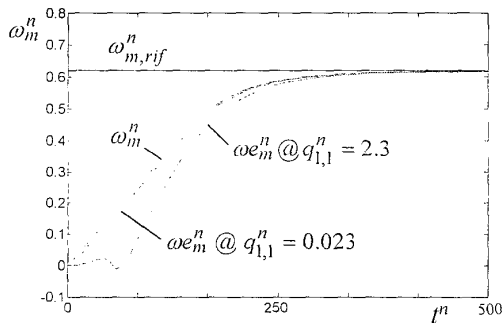


Fig. 2 – Speed at different $q_{1,1}^n$ (sensed system)

The control loops are then closed on the estimated speed and position quantities both in the computer simulations and in the laboratory prototype. The normalised settling time t_{stl}^n of the speed step response

has been used to evaluate the EKF closed loop behaviour at different $q_{1,1}^n$. Results at working point $\Omega^n = 0.25$, $T^n = 0.21$ are reported in Fig. 3, with $q_{3,3}^n = 0.46 \cdot 10^{-3}$, $q_{4,4}^n = 0.1$.

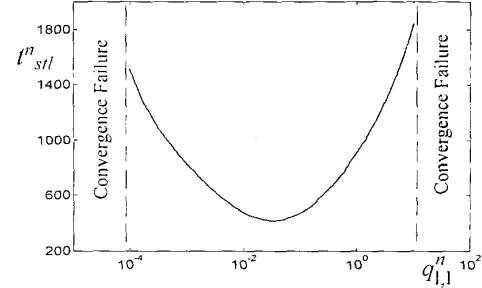


Fig. 3 – Settling time vs. $q_{1,1}^n = q_{2,2}^n$ ($\Omega^n = 0.25$, $T^n = 0.21$)

The minimum settling time is obtained for $q_{1,1}^n = 0.023$, for which the speed vs. time behaviour is similar to a first order response. For higher values of $q_{1,1}^n$ a marked overshoot appears. The delayed response of the filter corrupts the closed loop response up to instability, experienced for $q_{1,1}^n > 10$. Measurements have been carried out at low speed and load torque, where it is experienced a greater EKF sensibility to $\mathbf{Q}^n$ elements variation.

Opposite to $q_{1,1}^n$ and $q_{2,2}^n$, the effects of covariance elements $q_{3,3}^n$ and $q_{4,4}^n$ are more evident in the EKF closed loop system. Fig.4 shows the settling time as function of $q_{3,3}^n$, maintaining $q_{1,1}^n = 0.023$ and $q_{4,4}^n = 0.1$.

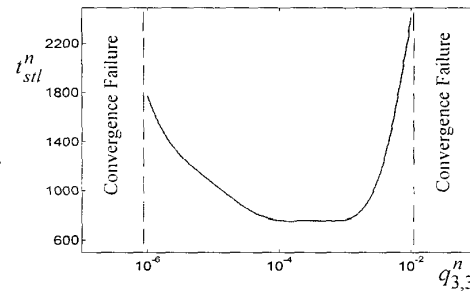


Fig. 4 – Settling time vs. $q_{3,3}^n$ ($\Omega^n = 0.375$, $T^n = 0.56$)

Small values of $q_{3,3}^n$ cause instability or swinging, while greater values involve strongly noisy signals but not convergence failure. Above $q_{3,3}^n = 0.01$ instability arises. There is anyway a quite broad range in which similar drive performance can be obtained.

It has been found that the element that most influences the EKF convergence is $q_{4,4}^n$; values in the range from 0.01 to 0.5 assure filter convergence and a good dynamics. Fig. 5 reports the settling time as function of $q_{4,4}^n$, maintaining $q_{1,1}^n = 0.023$ and $q_{3,3}^n = 0.46 \cdot 10^{-3}$. The convergence failure appears around $q_{4,4}^n = 0.5$.

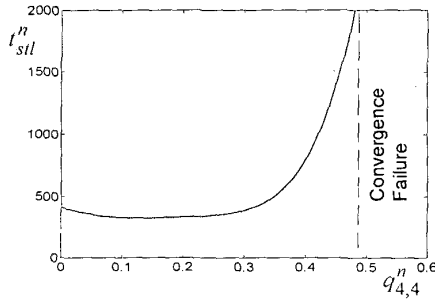


Fig. 5 - Settling time vs. $q_{4,4}^n$ ($\Omega^n=0.25$, $T^n=0.21$)

From these considerations it follows that the normalized Q^n matrix that yields the best results on the laboratory prototype is

$$Q^n = \begin{bmatrix} 0.023 & 0 & 0 & 0 \\ 0 & 0.023 & 0 & 0 \\ 0 & 0 & 0.46 \cdot 10^{-3} & 0 \\ 0 & 0 & 0 & 0.1 \end{bmatrix} \quad (11)$$

Matrix R^n concerns the measurements noise. Both simulation and experimental tests on the prototype point out that values in the range

$$r_{1,1}^n = r_{2,2}^n \in [0.2 \times 10^{-3}, 5] \quad (12)$$

do not influence significantly the EKF behaviour. Extremely high values of R^n increase the convergence time up to instability. An alternative guideline is to fix $R^n=I$, being I the identity matrix. For the system used in the experiments, it yields a normalized matrix R^n given by

$$R^n = \begin{bmatrix} 0.023 & 0 \\ 0 & 0.023 \end{bmatrix} \quad (13)$$

The condition $R^n=I$ allows a remarkable savings in the recursive calculation of the estimate error covariance matrix $P_{k|k}$. In details, for $R^n=I$, it is easy to prove that the following equivalences hold for the innovation step:

$$p_{1,1}^n = kf_{1,1}^n \quad p_{2,2}^n = kf_{2,2}^n \quad p_{1,2}^n = kf_{1,2}^n \quad (14)$$

and $p_{2,1}^n = p_{1,2}^n$. Eqn. (14) reduces of 30% the number of sums and of 20% the number of product operations of the EKF algorithm.

As mentioned before, the initial guess for P_0 slightly influences the EKF behaviour. The following normalised matrix has been used in this work:

$$P_0^n = \begin{bmatrix} 0.04 & 0 & 0 & 0 \\ 0 & 0.04 & 0 & 0 \\ 0 & 0 & 71 \times 10^{-6} & 0 \\ 0 & 0 & 0 & 10 \end{bmatrix} \quad (15)$$

Fig.6 shows the actual and estimated speed of the EKF sensorless prototype with the above calculated covariance matrices.

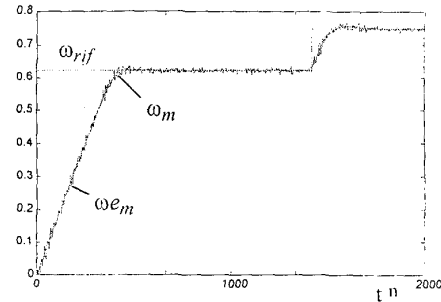


Fig. 6 - Speed response with Q^n , R^n , P_0^n

Experimental results perfectly match simulation outcomes, as a mark of the accuracy of this research.

5. Bartlett Test

A refinement of the design of the covariance matrices of the EKF algorithm can be obtained by a particular application of the Bartlett test. Let's define the steady state current prediction error as the difference between measured and predicted values of phase currents.

According to Kalman's theory, if exists a matrix Q that makes that prediction error to be a white stochastic process, then that Q makes the predictive algorithm reported in Table 2 to be optimal.

In the Bartlett test, the ideal integral of the prediction error spectrum density is a straight line with a slope proportional to the current variance. The degree of whiteness of the actual process can be evaluated as the displacement of the real integral with respect to the ideal line. As an example, Fig. 7 shows the Bartlett test performed on current ie_α^n during the on the field tuning of the Q^n matrix element $q_{1,1}^n$.

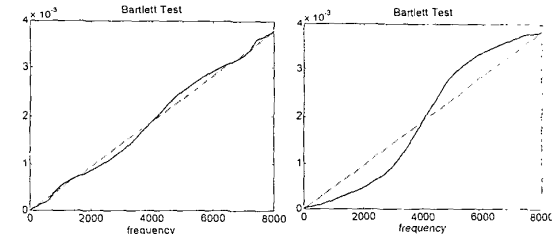


Fig.7 - Bartlett test on ie_α^n during the tuning of Q^n matrix (left: $q_{1,1}^n=0.023$, right $q_{1,1}^n=1.15$)

The Bartlett-optimal value $q_{1,1}^n=0.023$ is confirmed by the quasi-linear behaviour of the integral of the spectrum density, that is evidence of the whiteness of the current prediction error. Bartlett optimal Q^n matrix varies with the working point of the motor. Experiments have shown that $\Omega^n=0.2$ and $T^n=0.32$ is the steady state working point with the best sensibility to the Bartlett test.

On the other hand, it has been experienced that fine tunings performed far from the mentioned working point did not assure good dynamic performance in the whole operating range of the drive.

Conclusions

It has been presented a novel procedure for the tuning of covariance matrices in Extended Kalman Filter based PMSM drives.

The key-feature is the combined normalisation of both the controlled system model and the EKF algorithm. The first result is a proposal of normalised covariance matrices that should roughly fit for most of standard PMSM drives, overcoming the pitfall of a trial-and-error tuning. Secondly, a procedure for the fine-tuning based on a whiteness test is proposed.

Experimental tests, performed on working prototypes, have confirmed the validity of the procedures.

List of Symbols

x	System state vector
$\hat{x}_e$	Estimated state vector
u	System input vector
y	System output vector
B	Linear system input-state matrix
$f(x,u)$	System state functional
$F(x(t)) = \frac{\partial f}{\partial x} \Big _{x=x(t)}$	Jacobian of system state functional
H	System output matrix
K	Kalman gain matrix
P	State error covariance matrix
Q	Model noise covariance matrix
R	Measurement noise covariance matrix
i	Stator current vector
$\hat{i}_e$	Estimated stator current vector
R_s	Stator resistance
L_s	Stator inductance
Λ_{mg}	PM flux linkage
T_c	Sampling period
p	Motor pole pairs
ω, θ	Electrical rotor speed, position
$\hat{\omega}_e, \hat{\theta}_e$	Estimated electrical speed, position
ω_m	Mechanical speed
$\hat{\omega}_m$	Estimated mechanical speed
Subscripts, superscripts	
d, q	Synchronous frame quantities
α, β	Stationary frame quantities
k	Sampling index
$k k-1$	Predicted estimate
$k k$	Optimal estimate
0	Initial value
t	Transposed matrix
ref	Reference quantity
B	Base quantity
n	Normalised quantity

Bold characters are used for vectors and matrices. Symbol $\langle \rangle$ is used to indicate mean value in the interval from t_k to t_{k+1} .

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Appendix – PMSM Motor Data

The sensorless PMSM drive described in the paper has been implemented on a TMS320C31 TI DSP. Table 4 reports the data of the motor used in the experiments.

Table 4. PMSM Motor Data

Quantity	Value	Unit
Nominal torque T_B	2.8	Nm
Nominal speed Ω_B	420	rad/s
Nominal power P_B	0.909	KW
Max phase voltage U_B	120	V _{peak}
Nominal current	5	A _{rms}
Poles pairs	4	
R_s (@ 50°C)	1.5	Ω
L_s	3.5	mH
PM Flux linkage Λ_{mg}	0.066	Vs/rad